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Third-year CS student with an eye on real-time graphics and GPU programming (Rust + WebGPU, C++ + Vulkan), with two software engineering internships shipping and tuning production systems.

Education

University of California, Santa Cruz

Santa Cruz, CA

Expected Jun 2028

B.S. in Computer Science | GPA: 3.6/4.0

Honors: Merit Scholarship, Dean's Honors

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Systems Programming in C, Linear Algebra, Vector Calculus, Probability & Statistics

Activities: UC Santa Cruz Game Design & Art Club

Experience

Paid Software Engineering Intern

Flickr

Jun 2023 - Aug 2023

- Cut compute cost 92% and raised throughput 39% on Wolverine, a Python AWS Lambda photo-repair service running over a library of ~10 billion photos, by tuning memory and thread allocation to the cheapest stable configuration.
- Profiled dozens of Lambda configurations on controlled batches of real jobs in Splunk, modeling cost per job, throughput, and tail latency to find the cost/performance point AWS docs do not surface.
- Recovered AWS's undocumented vCPU-to-memory scaling ratio by measuring CPU behavior across memory settings and fitting a linear regression, enabling accurate per-job cost estimates.

Paid Software Engineering Intern

SmugMug

Jun 2022 - Aug 2022

- Owned a user-facing gallery stats feature end to end (PHP, React, Next.js), adding daily refresh scheduling and clearer labels and empty states, shipped through PR review and QA.
- Wrote unit tests and backend changes supporting a release-critical PHP 8.1 upgrade, pairing with senior engineers to land it on schedule.
- Traced and fixed a broken checkout-flow link buried in a large monorepo, shipping the fix through a multi-stage review and QA pipeline.

Projects

ray-vox: Ray-traced voxel renderer (in progress)

Rust, WebGPU, Data Structures, Optimization

2026 - Present

- Built the core sparse-voxel data structure for a from-scratch ray-traced renderer in Rust and WebGPU, storing an 84 MB scene in 16 MB, smaller than zstd's most aggressive setting (19 MB) while staying directly GPU-traversable.
- Eliminated all per-node child pointers using per-node bitmasks and bit-counting, and packed node offsets into a single 32-bit word, cutting memory while keeping ray traversal branch-light for GPU warps.
- Designed the format as a GPU acceleration structure with built-in level-of-detail and matching on-disk and in-GPU-memory layout, so uploads are near-zero-conversion copies.

Crowd Surfers: Real-time 3D game

Godot, Shaders, Lighting, Architecture

2025 - 2026

- Led a mid-project migration from a faked-depth sprite system to true 3D with an angled orthographic camera, building a working prototype that convinced a 100-person student team to adopt the rewrite.
- Built a 3D occlusion-based transparency shader that fades buildings as the player skates behind them, using a camera-to-player frustum test plus dithered alpha to fit the alpha-cut asset pipeline.
- Added real-time shadows, dynamic lighting, and camera feel (speed-based FOV, screen shake, look-ahead), all smoothed with interpolation.

OKLab RGB Cube Bijection

Rust, OKLab, Algorithms

2026

- Arranged all 16.7 million RGB colors into one 4096x4096 image, each color used exactly once and reading as a smooth gradient, via a recursive cube-subdivision permutation optimized against a blurred copy in OKLab perceptual color space.
- Built a palette generator producing perceptually uniform N-color palettes by iteratively removing the least-distinct color in OKLab, surfacing a hard limit of linear RGB near one million colors.

FRC Swerve Drive

Java, Control Theory, Computer Vision

2023 - 2024

- Built a coaxial swerve drivetrain from scratch (custom CNC chassis, electronics, ~2,000 lines of Java) with no prior team experience, leading an off-season team to a working prototype in one month, ~4 months ahead of schedule.
- Ran per-module PID and feedforward at a 1 kHz control loop and derived current limits from robot mass and tire friction to maximize grip without slip or brownout, after self-teaching graduate-level control theory.
- Fused wheel encoders, gyro, and AprilTag vision through a Kalman-filter pose estimator for sub-centimeter localization robust to wheel slip, and simulated the full robot (torque curves, inertia, battery, brownouts) on the exact production code.

Skills

Languages: Rust, C++, Python, Java, TypeScript / JavaScript, PHP, GLSL / WGSL

Graphics & GPU: Vulkan, WebGPU, Ray Tracing, Real-Time Rendering, Shaders, Level-of-Detail

Tools & Cloud: Git, Linux, AWS (EC2, Lambda, Graviton), Docker, OpenCV, Godot, React / Next.js

Concepts: Performance Optimization, GPU Data Structures, Control Theory, Computer Vision, Pose Estimation